Storm-to-Outage: Exploratory Data Analysis of the ML-Ready Dataset

DSCI 4350 capstone · Georgia State University · Fall 2026
Question: **given a storm, what is the probability that a US county loses power?**
Dataset version: Hugging Face `DSCI-Capstone/county-outage-data` @ `5c01325` · code: notebooks 01–04 (build), 06 (this EDA)

How to read this report. Section 1 lists the findings. Sections 2–7 give the evidence for each. Section 8 summarises the earlier EDA on the cleaned outage and storm tables, which shaped the target and unit. The **dataset card** (origin, lineage, cleaning steps, feature dictionary, statistics, limitations) is attached at the end.

Ground rules. Feature–target relationships use **training rows only (2017–2023)**, so the validation (2024) and test (2025) years can't influence modeling choices. Distributions are compared across all three splits using **features only**. The 2025 labels are never read.

1. Key findings

1. **The ML-ready table has 271,644 county × storm-episode rows** (train 208,482 · validation 31,064 · test 32,098), 36 features known when the storm starts, and a 14.1% positive rate in training (≥5% of customers out within 24 h).
2. **The positive rate rises over time**, from 11–12% in 2017–2019 to 15–17% from 2020 and 16.5% in validation. EAGLE-I's growing coverage explains part of this. **Models need a calibration check on 2024.**
3. **County history is the strongest single signal** (ROC-AUC 0.74 alone). The outage rate climbs from 1.5% in the lowest-history decile to 37% in the highest. **Where a storm hits matters as much as what the storm is.**
4. **Storm type separates risk sharply.** Tropical systems lead to an outage 62.5% of the time, ice storms 33%, tornadoes 29% and thunderstorm wind 25%, against 3–8% for winter, cold, wildfire and other types.
5. **Counties already out stay out.** When ≥5% of customers are out the hour before the storm, 89% of rows are positive (1.2% of rows). Any pre-storm outage at all raises the rate above the base.
6. **How NOAA located the storm matters.** County-coded storms lead to an outage 17.9% of the time; zone-coded storms only 9.9%. That's because a zone report is spread over every county in the zone.
7. **Missing values carry information.** No wind measurement means no severe-wind report, and those rows are positive 8.6% of the time vs 22.1% with one. Missing values should be modeled, not filled blindly.
8. **Small-to-mid counties are most at risk** (~18%); the largest counties drop to ~5%. Part of this is real (urban grids) and part is the threshold (5% of a small county is few customers).
9. **Four feature pairs are near-duplicates** ($|\rho| \geq 0.93$). That's harmless for trees but matters for logistic regression.
10. **Features are stable across splits** (PSI < 0.1) **except** `dqi` (PSI 3.7). `dqi` is a year-mapped quality score, so it acts as a year marker and should be dropped or handled with care.

2. The dataset at a glance

Split	Years	Rows	Counties	Storm episodes	Positive rate
Train	2017–2023	208,482	3,073	54,580	14.1%
Validation	2024	31,064	3,023	7,734	16.5%
Test	2025	32,098	3,014	7,866	held out

What a row is. One NOAA storm episode (`episode_id`) in one county (`fips`). All of that episode's reports in the county are summarised in one row. The clock starts at `t0`, the first report there.

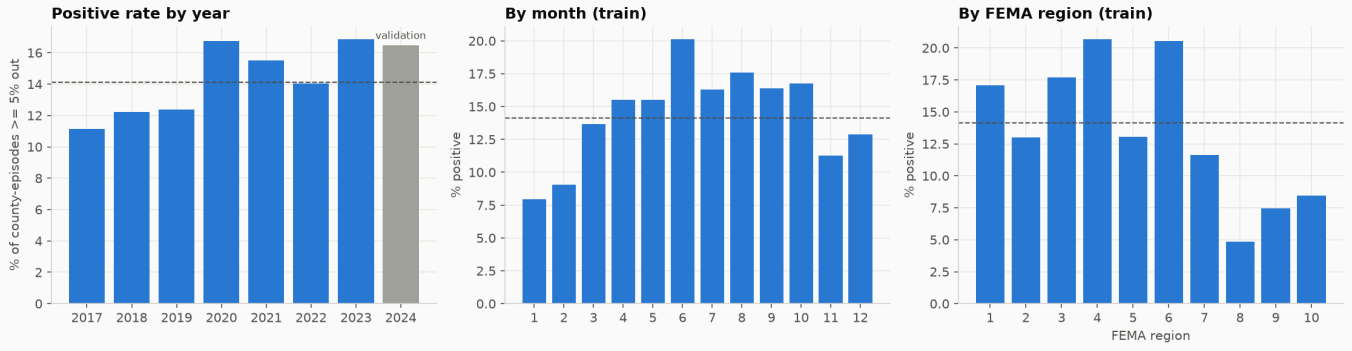
Target. `target = 1` if the peak share of the county's customers without power, over the 24 hours from the start hour, reaches 5%.

Features (36), in five groups:

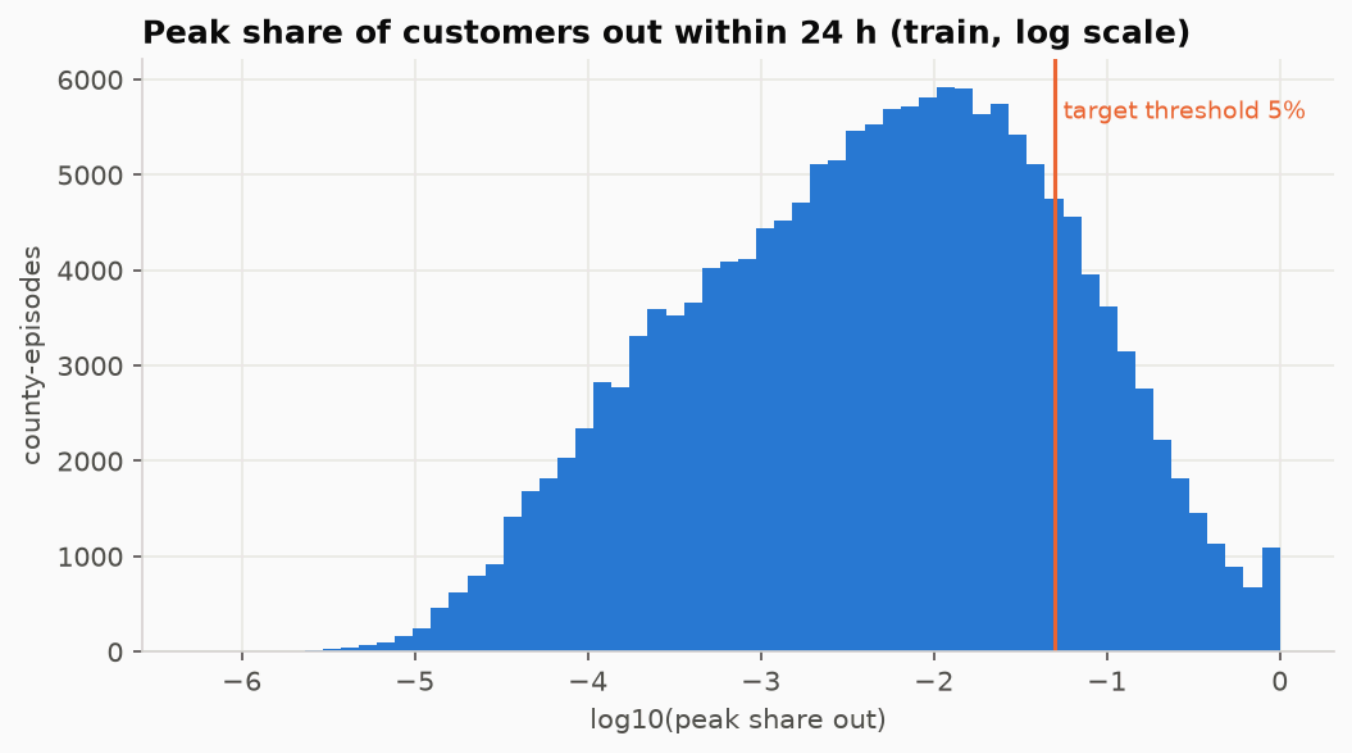
Group	Count	Examples
Storm (whole episode in the county)	25	reports per storm family, county- vs zone-coded reports, max wind and hail, hours active, storm-system size
Time	2	month, local hour of the start
County	4	customers served, population, EAGLE-I quality scores
Before the storm	2	worst share out in the previous 24 h, share out in the hour before
County history (training labels only)	3	past episodes, past outage rate, past mean peak

Operations applied (details in the dataset card, §4–5): raw outage readings were cleaned and given a denominator, then joined to storm reports expanded to counties. The rows were collapsed to county × episode, and only rows where EAGLE-I was observing the county were kept. **643,757 storm × county rows became 271,644 modeling rows.**

3. The target: how often, when and where



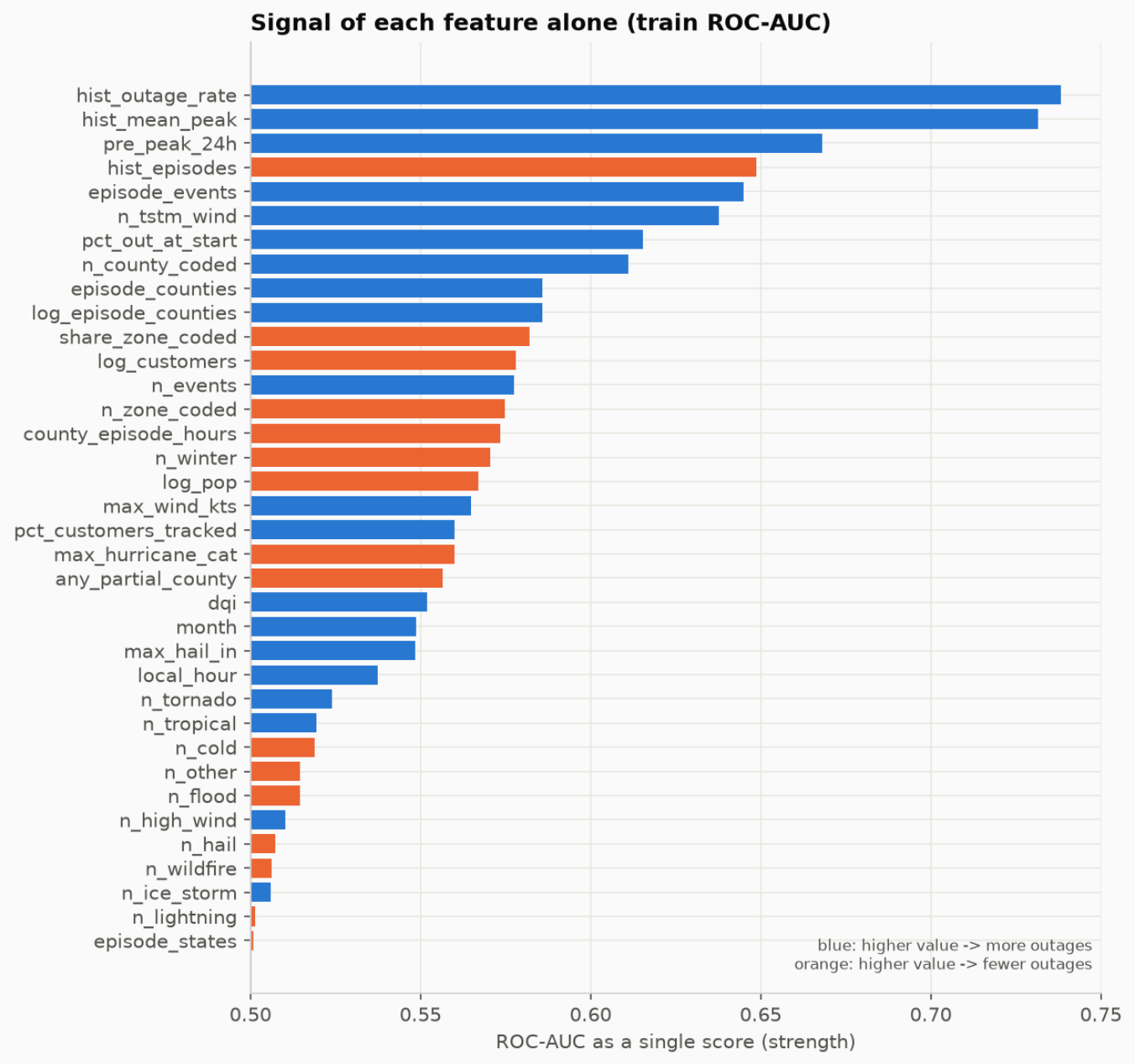
- **Rising over time:** 11.2% in 2017, 16.7% in 2020 and 16.8% in 2023. Validation (2024) sits at 16.5%, above the training average of 14.1% (dashed line).
- **Seasonal:** 8% in January and 20% in June. Summer convective storms do more damage per episode than winter systems.
- **Regional:** FEMA regions 4 (Southeast) and 6 (South Central) are at ~21%; region 8 (Mountain) is at 4.9%. The highest states are LA (37.5%), MS (28.9%), GA (27.4%), ME (25.5%) and WV (25.2%); the lowest are CO (3.1%), WY (3.6%) and DC (1.2%).



- The label-defining peak has a long tail. **22% of county-episodes have nobody out at all**, 32% reach 1%, 14.1% reach 5% (the target), 8.3% reach 10% and 1.2% reach 50%. **Among positives, the median peak is 11.9%**, so positives are well past the threshold, not borderline.

4. What predicts outages: feature–target relationships

4.1 Signal of each feature on its own

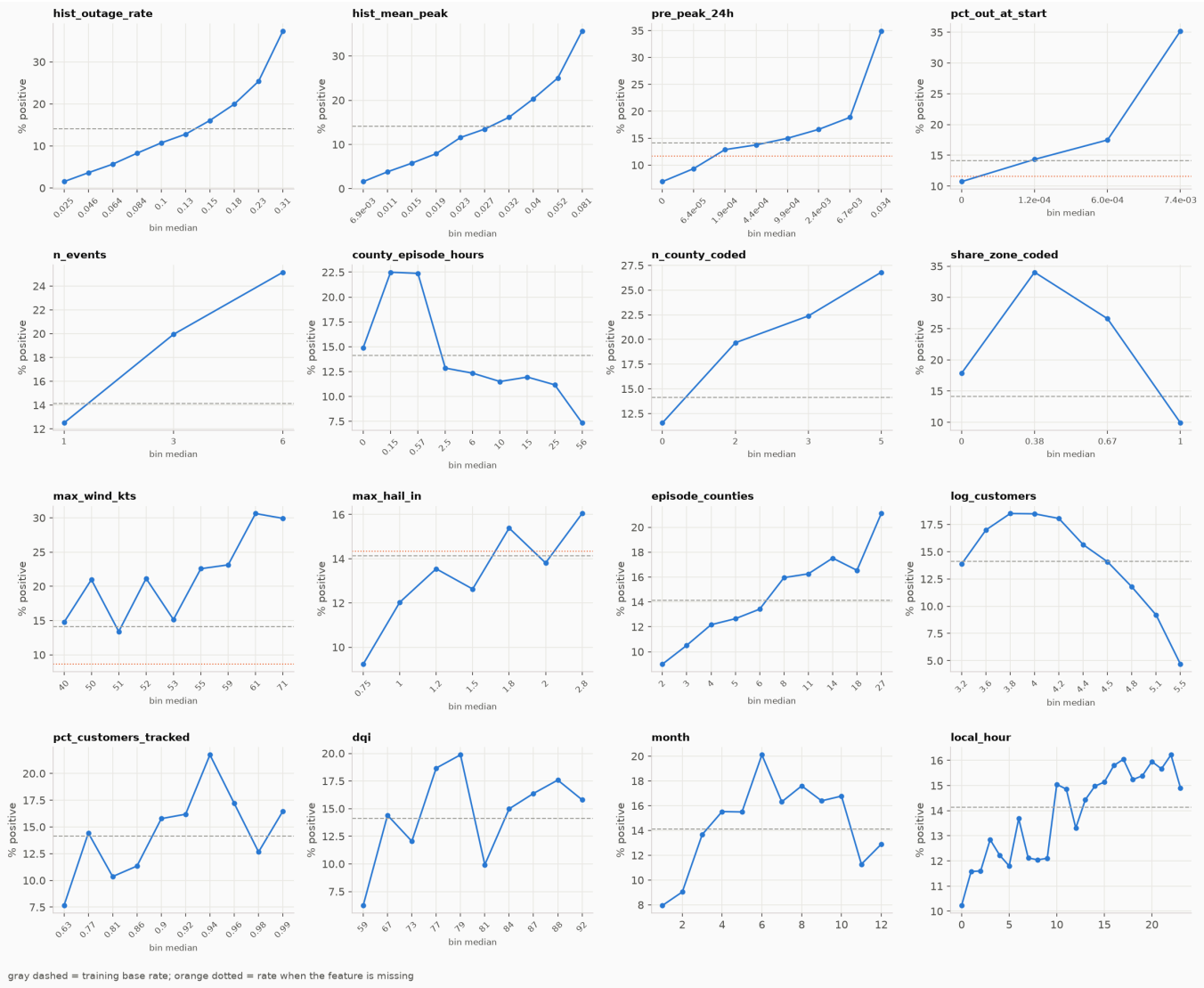


Each bar shows how well one feature, used alone as a score, ranks positives above negatives (ROC-AUC; 0.5 = no signal).

- **History leads:** past outage rate 0.74 and past mean peak 0.73.
- **Then pre-storm conditions** (worst share out in the previous 24 h: 0.67), **storm-system size** (reports in the episode: 0.65), **thunderstorm wind reports** (0.64) and **county-coded reports** (0.61).
- **Inverse signals (orange):** more past episodes, more zone coding, bigger counties and longer activity all mean *lower* risk. Long-lasting, zone-coded winter systems are common but rarely knock out 5% of a county.

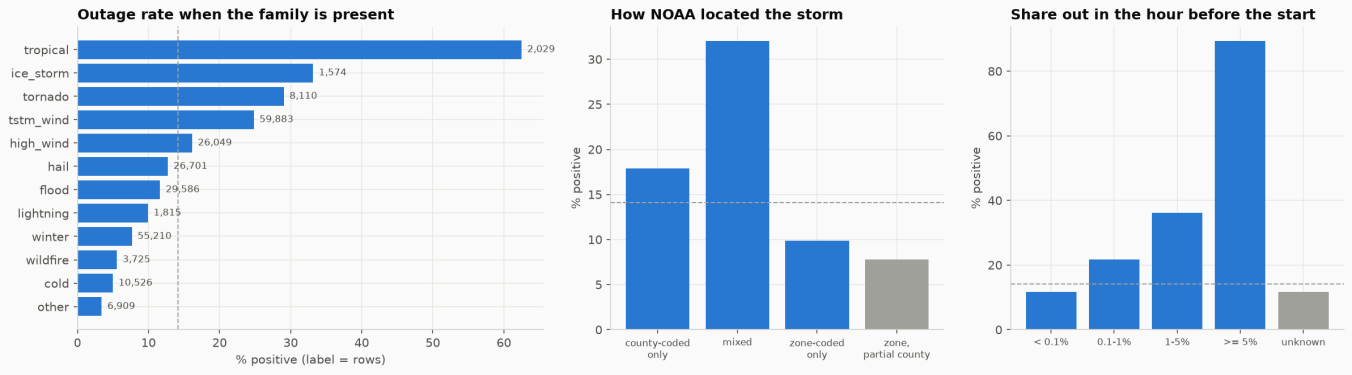
- No single feature is near the 0.80 ROC-AUC of the simple "storm family × county history" rule, so **combining features is where a model earns its keep**.

4.2 Outage rate across each feature's range



- **History and pre-storm features rise smoothly** and almost linearly. These are good inputs for both logistic regression and boosting.
- **Max wind:** rows with a measured gust of 61+ knots are positive about 30% of the time. Rows **without** a wind report (orange line) are positive only 8.6% of the time.
- **County size has a hump:** small-to-mid counties are highest, the largest are lowest. A linear model needs a transformation to capture this; trees capture it directly.
- **Activity duration is non-monotonic:** short, intense county episodes (under an hour) are most dangerous (22%). Multi-day episodes fall to 7%.
- **Hour of day:** risk climbs from ~10% at midnight to ~16% in the afternoon and evening, following the convective-storm cycle.
- **`dqi` and `pct_customers_tracked` look noisy.** They are region- and state-level quality scores, not storm drivers.

4.3 Storm family, storm location coding and pre-storm outages



Storm family present	Rows	% positive		Pre-storm share out	Rows	% positive
Tropical	2,029	**62.5%**		< 0.1%	176,457	11.7%
Ice storm	1,574	33.2%		0.1–1%	16,697	21.7%
Tornado	8,110	29.0%		1–5%	5,807	36.1%
Thunderstorm wind	59,883	24.8%		≥ 5%	2,596	**89.4%**
High wind	26,049	16.2%		unknown	6,925	11.6%
Hail / flood	26–30 K	12–13%				
Winter / cold / wildfire / other	3.7–55 K	3–8%				

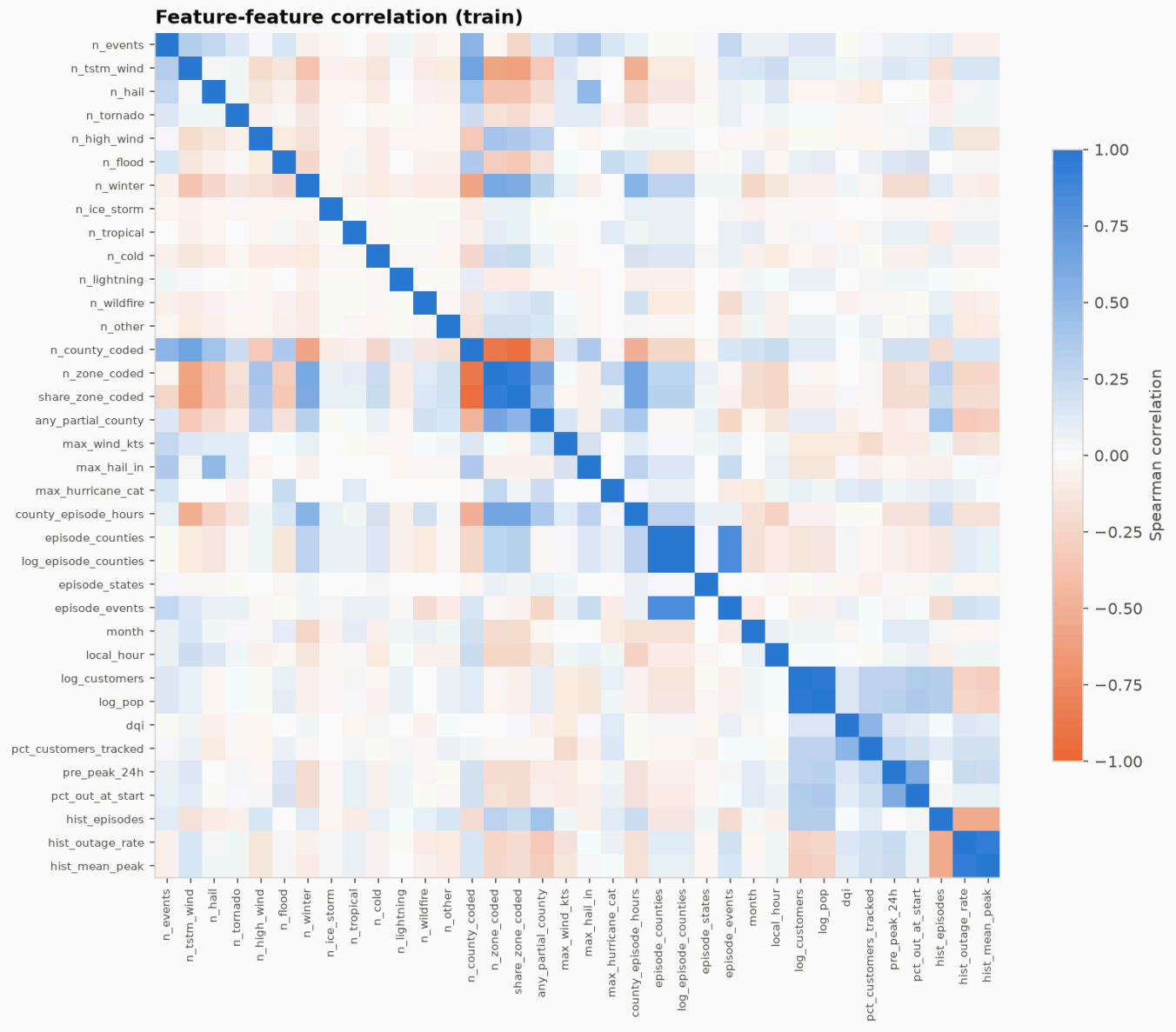
- **Storm family is a strong but blunt feature.** County history alone spans 1.5% to 37% (§4.2), so the same storm type means very different risk in different counties. That's why the reference rule multiplies the two.
- **Zone-coded only: 9.9%. County-coded only: 17.9%. Mixed episodes: 32%.** Mixed episodes are large systems with both kinds of report. Partial-county zones are the weakest (7.8%).
- **Already-out counties (≥5% before the start)** are 1.2% of rows but nearly always positive. The target keeps them on purpose (user decision), and `pct_out_at_start` lets the model recognise them.

5. Missing values carry information

Feature	Missing	% positive when missing	% positive when present	Why it's missing
`max_hurricane_cat`	99.9%	14.1%	90.2%	Only hurricanes have a category
`max_hail_in`	87.2%	14.3%	12.7%	No hail report in the episode
`max_wind_kts`	59.0%	**8.6%**	**22.1%**	No report with a wind measurement
`pre_peak_24h`, `pct_out_at_start`	3.3%	11.6%	14.2%	Day before not observed by EAGLE-I
`log_pop`	0.4%	11.3%	14.1%	Connecticut's 8 old counties have no 2023 Census population

Implication: keep "missing" as information. `HistGradientBoostingClassifier` handles it natively; logistic regression gets a missing-flag per feature plus a neutral fill.

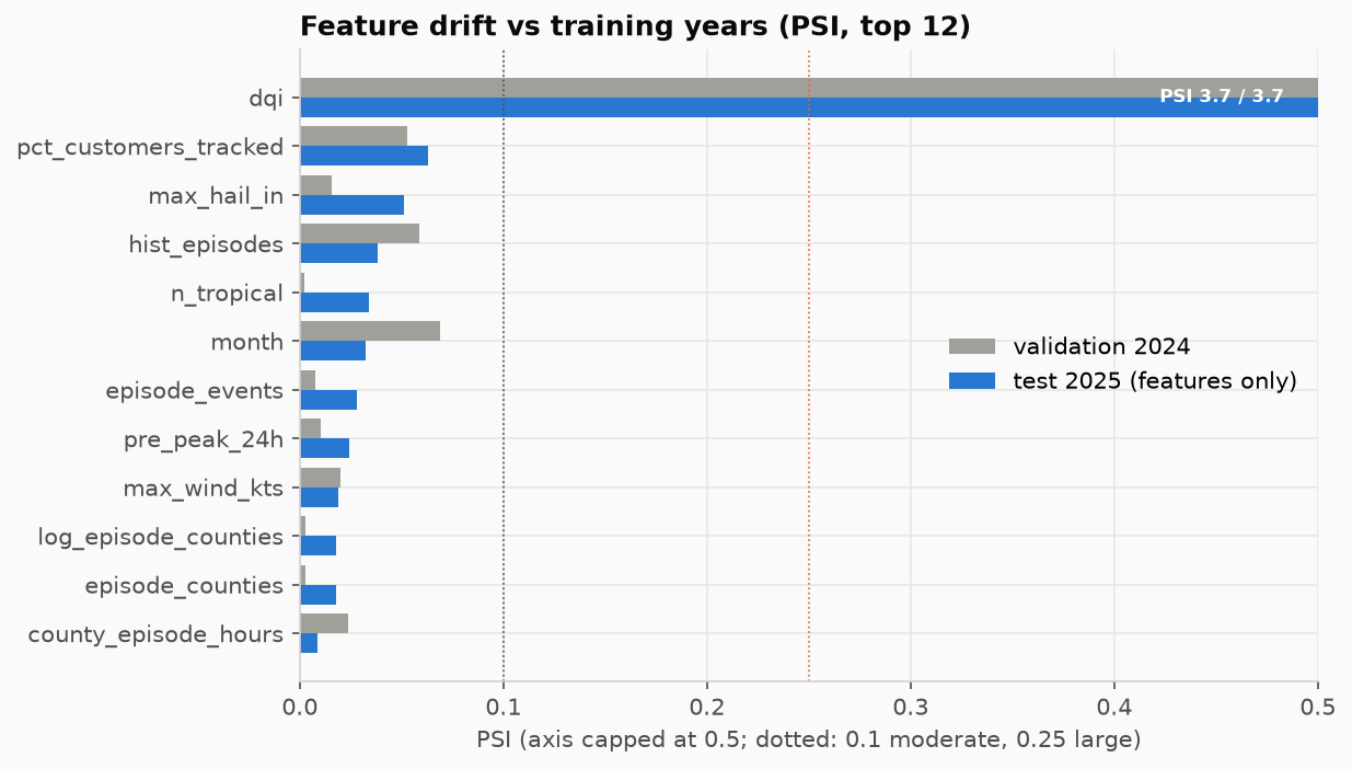
6. Redundant features



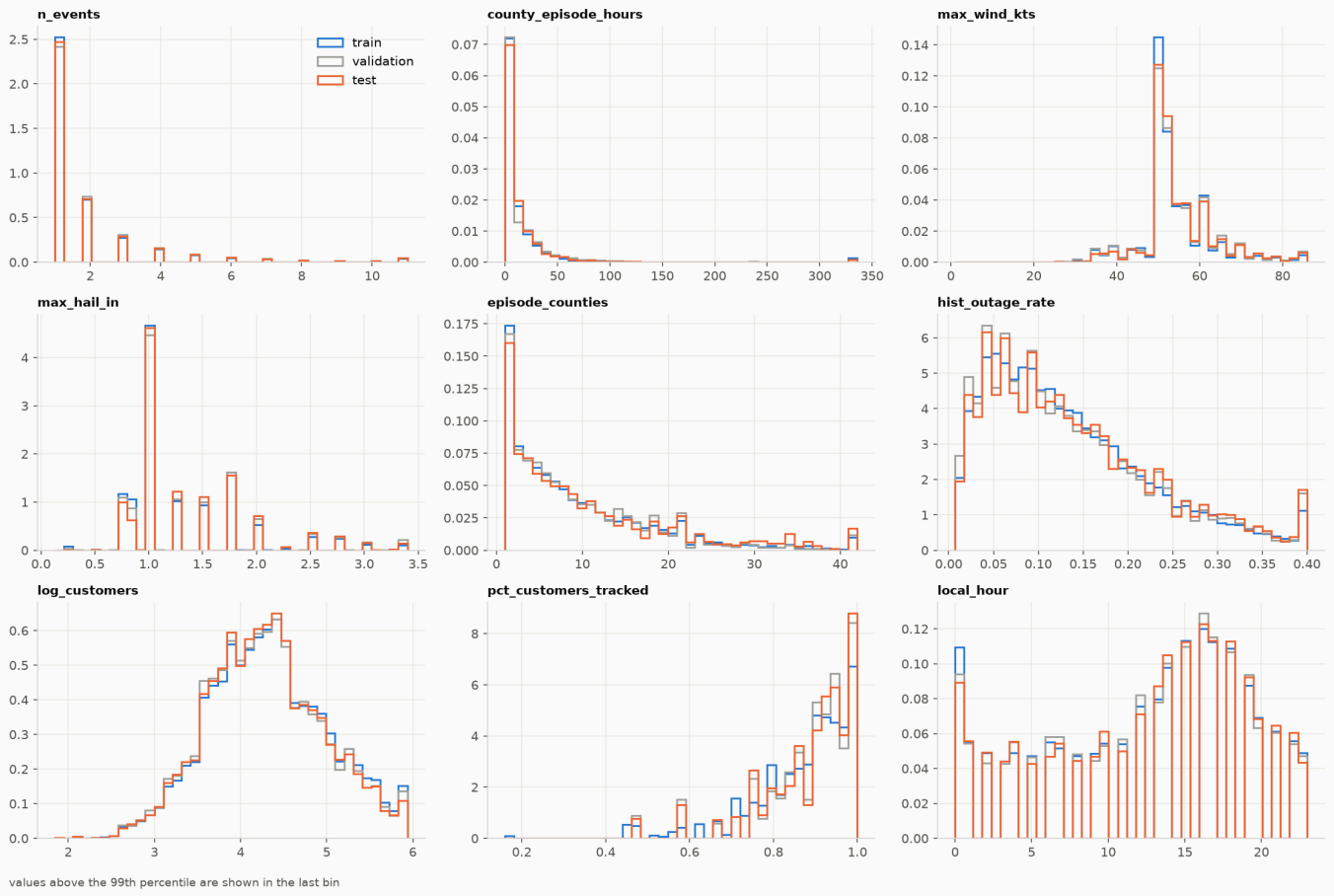
Feature pair	Spearman ρ	Takeaway
`episode_counties` · `log_episode_counties`	1.00	Same ranking (log transform); keep one for linear models
`log_customers` · `log_pop`	0.98	Customers ≈ 0.5 × population almost everywhere
`n_zone_coded` · `share_zone_coded`	0.95	Most episodes are all-zone or all-county
`hist_outage_rate` · `hist_mean_peak`	0.94	Two views of the same county history
`n_county_coded` · `share_zone_coded`	−0.93	Mirror image of the pair above

Trees are unaffected. For logistic regression, these pairs inflate coefficient variance, so use regularisation (or drop one of each pair) and don't read coefficients one by one.

7. Stability across splits (drift)



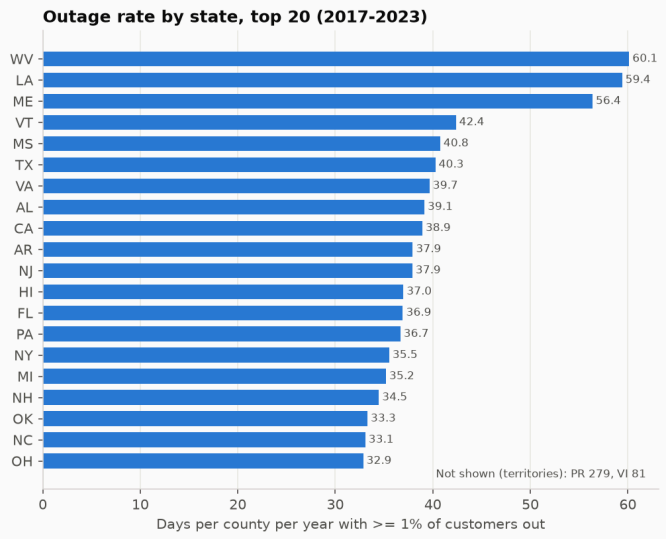
- **Almost every feature is stable:** population stability index < 0.1 for validation and test.
- `dqi` **breaks (PSI 3.7):** the EAGLE-I quality index only exists for 2018–2022, so every 2023–2025 row gets the 2022 regional value. In training it partly encodes the year, and in validation and test it is nearly constant. **Recommendation: leave** `dqi` **out of the baseline models** (`pct_customers_tracked` has the same mapping but drifts only mildly, PSI 0.06).
- `hist_episodes` is ~17% larger in validation and test (7 years of history instead of 6), a known effect of the leave-one-year-out design. It's small (PSI < 0.06), but worth knowing.
- **2025 has almost no tropical activity** (9 rows), so tropical performance can only be judged on validation.

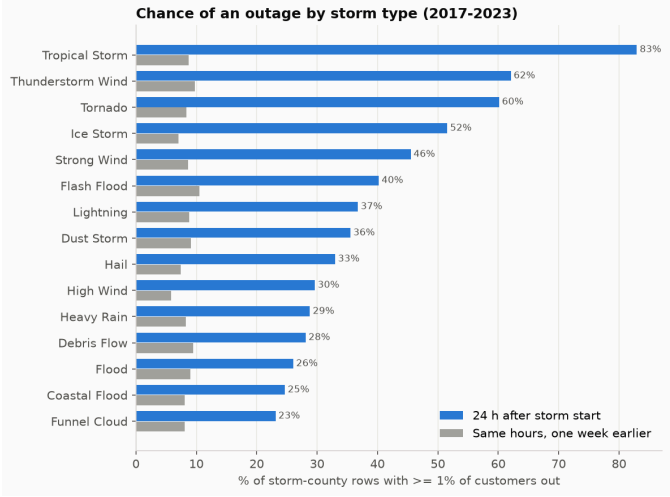


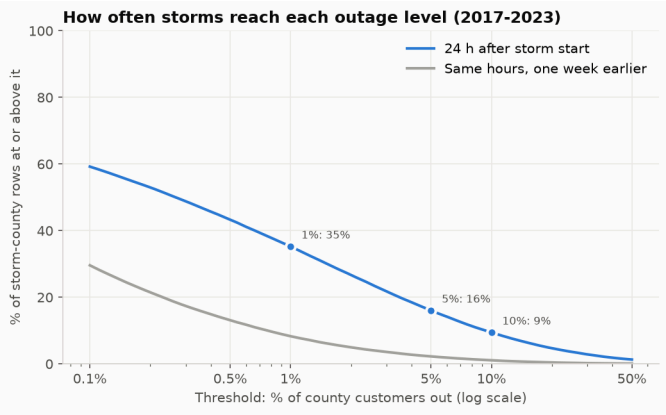
The distributions of the main features overlap closely across train, validation and test. Wind reports cluster at 50 knots, NOAA's severe-wind criterion, so wind speed carries less information than its range suggests.

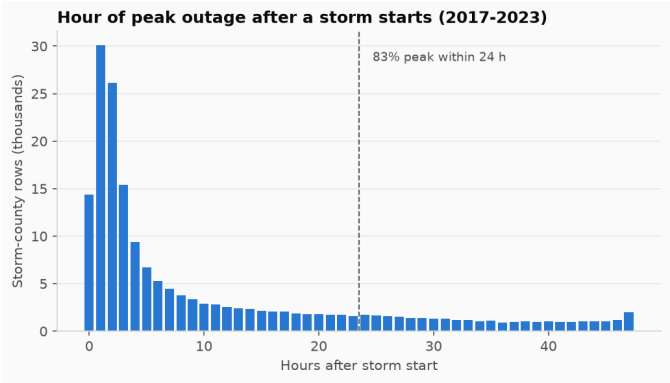
8. Context: outage and storm patterns in the cleaned data (2017–2023)

This EDA ran on the cleaned outage and storm tables *before* the ML-ready table existed. It set the target (≥5% within 24 h), the unit (county × episode) and the triggers. A "significant outage" there is ≥1% out within 24 h of a storm report.









- **Storms multiply outage risk:** 35% of storm × county rows have a ≥1% outage within 24 h, against 8% in the same county and hours one week earlier. At the ≥5% level, the ratio is ~8×.
- **Outages peak fast:** median 4 h after the storm starts, and 83% within 24 h. That's why the window is 24 h.
- **Thresholds:** ≥1% gives 36% positives but more background noise; **≥5% (17%) is a cleaner storm signal**. The project uses 5%.
- **Most outages have no storm record.** Only ~20% of significant outage county-days have a NOAA storm within ±1 day, so the model covers storm-linked outages only.
- **Drought and heat barely differ from a normal week**, so they were dropped as triggers.

9. What this means for modeling

Finding	Modeling choice
Rising positive rate, partly from coverage	Check calibration on 2024; time-based splits only
History and pre-storm outages are the strongest signals	Keep the leave-one-year-out history; no extra leakage guards needed beyond `retro_*`
Non-linear effects (county size hump, duration, wind threshold)	Gradient boosting is the natural main model; logistic regression needs transforms
Missing values carry information	Native missing handling (boosting) or missing flags (logistic regression)
Near-duplicate feature pairs	Regularise logistic regression; drop one of each pair if needed
`dqi` drifts by construction	Recommended: exclude it from the baselines
Rows within an episode are dependent	Bootstrap by `episode_id` when comparing models
Bar to beat	Storm family rate × county history: 2024 PR-AUC 0.469, ROC-AUC 0.80 (see `reports/success_criteria.md`)

Attachment: dataset card. Also available as `reports/dataset_card.md` in the repository and as the Hugging Face dataset README.

Dataset Card: Storm-to-Outage County Episodes (ML-ready)

Name	Storm-to-Outage County Episodes, 2017–2025
Version	Hugging Face dataset commit `5c01325` · GitHub commit `02c4471` (notebook 04) · built 2026-10-25
Hosted at	huggingface.co/datasets/DSCI-Capstone/county-outage-data (private; access through the `DSCI-Capstone` org)
Code	github.com/Chumpz0/Capstone-Power-Outage-Prediction, notebooks 01–04 build it, 06 profiles it
Maintainer	Christopher Walker, DSCI 4350 capstone team (Georgia State University, Fall 2026)
License	CC BY 4.0 (inherits EAGLE-I's license; NOAA, NWS and Census inputs are public domain)

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1. Summary

One row per **US county hit by one NOAA storm episode**, 2017–2025, lower 48 states + DC. Each row describes the storm in that county, the county itself, outage conditions just before the storm, and the county's past outage history. The binary **target** says whether at least **5% of the county's electricity customers lost power within 24 hours** of the storm's first report there.

Intended use: train and evaluate models of *P(county outage | storm)* for **utility crew staging**, i.e. ranking which counties to pre-position crews in when a storm system arrives.

Not intended for: predicting outages without a NOAA storm record (most outages, see §7), counting customers affected, or real-time use without swapping the observed storm summary for a forecast.

2. Structure

File (HF path)	Rows	Columns	Years	Positive rate
`processed/model_table/train.parquet`	208,482	46	2017–2023	14.1%
`processed/model_table/validation.parquet`	31,064	46	2024	16.5%
`processed/model_table/test_features.parquet`	32,098	42	2025	not shown
`holdout/test_labels.parquet`	32,098	6	2025	**held out: open once, for the final evaluation**

- **Columns:** 6 identifiers, **36 model features**, `target`, and 3 `retro_*` columns that are only known after the storm (analysis only, **never model inputs**).
- **Keys:** `fips` (5-character text, keep leading zeros) + `episode_id` are unique across all splits.
- **Splits are by time** (year of the episode's first report in the county). The test year is touched once, at the end.

- **Coverage:** 3,073 counties, 54,580 storm episodes and 49 states/DC in train; 3,023 counties and 7,734 episodes in validation; 3,014 and 7,866 in test.

Upstream tables are in the same dataset: `data/` (raw inputs) and `processed/` (cleaned 15-minute and hourly outages, daily observation flags, cleaned storm events, storm × county, county lookup).

3. Origin

Source	What	Version used	License
EAGLE-I (Oak Ridge National Laboratory), figshare 24237376	County customers-out snapshots every 15 minutes, 2014–2025 (249.3 M rows) plus reference files: modeled customer counts (MCC), data quality index (DQI), coverage history	Yearly files combined into one CSV (13 GB)	CC BY 4.0
NOAA Storm Events (NCEI), csvfiles	Storm reports with type, time, location, magnitude, episode id, 2014–2025 (771 K events)	`StormEvents_details-ftp_v1.0_dYYYY_c2026MMDD` (files created 2026-03-23 to 2026-08-19)	Public domain
NWS zone–county correlation (weather.gov)	Which counties each forecast zone covers	23 dated versions (Oct 2014 – Apr 2026), combined into a dated crosswalk	Public domain
US Census	County population estimates (Vintage 2023); Connecticut planning region ↔ county crosswalk	`co-est2023-alldata`, `sub-est2023_9`	Public domain

4. Lineage

```
EAGLE-I CSV (249.3 M rows) --+
NOAA Storm Events (771 K) ---+-- 01 audit: types, nulls, coverage, leakage tags (no data changed)
NWS zone crosswalk ----------+
Census, MCC, DQI ------------+-- 02 prep
                                -> outages_15min (208.1 M) -> outages_hourly (61.4 M)
                                -> county_observed_daily (10.7 M county-days)
                                -> storm_events_clean (573 K) -> storm_county_events (643,757)
                                -> counties (3,248)
                                       |
04 modeling table <---------------------+
   643,757  storm x county rows
   629,256  lower 48 + DC
   558,402  drought / heat / excessive heat dropped
   298,398  collapsed to county x episode
   271,644  county observed on the start day and the window's end day (FINAL)
            train 208,482 | validation 31,064 | test 32,098
```

5. Cleaning and transformation steps

Outages (notebook 02)

1. Kept 2017–2025 (2014 starts on Nov 1, and county coverage only reaches ~92% from 2017). 221.4 M raw rows.
2. **Dropped 8.7 M null readings** (they are gaps, not zeros) and **4.8 M explicit zero rows** (zeros are only reported from 2022). A missing row means "0 or no data", so a separate **daily observation flag**

records whether EAGLE-I was watching the county (county reported that year, ≥90% of the day's snapshots collected nationally, readings not all null).
3. Snapped 20 K off-grid timestamps to the 15-minute grid.
4. **Denominator:** EAGLE-I's modeled customer count (MCC). `total_customers` from the raw file was dropped (100% null except 2024, where it is unreliable). Where MCC is broken, an estimate of Census population × 0.508 customers per person is used: **all 100 North Carolina counties** (MCC gives 0.17 customers per person vs 0.35–0.60 elsewhere), 9 counties with no MCC, and 21 counties below 0.1 or above 5 customers per person.
5. **Capped customers out at the denominator** (~0.05% of rows), flagged `over_mcc`, raw value kept.
6. Connecticut: 2025 planning-region rows converted back to the 8 old counties with population weights.
7. Built an hourly table (worst snapshot per county-hour).

Storms (notebook 02)

1. Dropped marine and waterspout events, and **33 leakage or unneeded columns** (damage, injuries, deaths, narratives, tornado scale and path size, which are only known afterwards).
2. Converted local times to **UTC**; fixed NOAA territory state codes.
3. Expanded **forecast-zone reports to counties** with the zone crosswalk version in force on the event date; flagged zones that cover only part of a county (`partial_county`).

Modeling table (notebook 04)

1. Scope: lower 48 + DC; Drought, Heat and Excessive Heat are not storm triggers.
2. **Unit:** all reports of one NOAA episode in one county collapse into one row; `t0` = first report there.
3. **Kept only county-episodes EAGLE-I observed** on the start day and the 24 h window's end day (drops 26,754 rows, mostly in 2017–2021 when coverage was lower).
4. **Target:** peak share of customers out over hours 0–23 from the start hour ≥ 5%. Hours with no outage row count as 0 (the county is known to be observed).
5. **Storm features** summarise the whole episode in the county: report counts by family, how reports were located, max wind/hail, duration, and storm-system size. This **observed summary stands in for a forecast**.
6. **Pre-storm features** from the 24 h before the start hour (missing when the day before wasn't observed).
7. **County history** from training-year labels only, shrunk toward the national rate with 20 pseudo-episodes. Training rows use the *other* training years (leave-one-year-out) so no row sees its own label; validation and test rows use all training years.
8. Test labels are written to a separate file.

6. Dataset statistics

Target

- 14.1% positive in train, rising over time: 11.2% (2017), 12.2%, 12.4%, 16.7% (2020), 15.5%, 14.0%, 16.8% (2023); validation 2024 16.5%.
- The label-defining peak: 22% of county-episodes have no customers out at all in the window, 32% reach 1%, 14.1% reach 5%, 8.3% reach 10%, 1.2% reach 50%. The median peak among positives is 11.9%.

Features (training rows; full table in `reports/ml_eda_tables/feature_dictionary.csv`, missing values by split included)

- A typical county-episode has **1 report** (65% of rows; mean 1.9, max 83) and is active in the county for a median of 4 hours (75th percentile 15 h). The whole storm system touches a median of 7 counties (mean 9.7).
- **71% of rows have no thunderstorm wind report and 87% no hail**; tropical, ice storm and lightning are present in ≤1%.
- **Half of all rows are zone-coded only** (101,801), half county-coded only (104,315), ~1% mixed.
- **Missing values:** `max_hurricane_cat` 99.9%, `max_hail_in` 87.2%, `max_wind_kts` 59.0% (empty when no report of that kind; the empty value carries information), `pre_peak_24h` and `pct_out_at_start` 3.3% (day before unobserved), `log_pop` 0.4% (Connecticut old counties). All other features are complete.

7. Known limitations and biases

- **Only storm-linked outages.** In 2017–2023 only ~20% of significant outage county-days have a NOAA storm record within ±1 day. The data says nothing about outages without one.
- **Observed storms stand in for forecasts.** Real forecast-time accuracy will be lower.
- **Coverage grows over time:** EAGLE-I watched 84–88% of county-episodes in 2017–2021 vs 92–98% in 2022–2025, and tracks more customers each year. This is part of why the positive rate rises; models trained on all years may under-predict 2024–2025.
- **`dqi` and `pct_customers_tracked` are year-mapped quality scores** (values exist for 2018–2022 only; 2023–2025 all use 2022). `dqi` shifts strongly between train and validation/test (PSI 3.7) and partly acts as a year marker. Use with care.
- **`hist_episodes` scale differs by split:** training rows count 6 other training years, validation/test rows count all 7 (mean 87 vs 102).
- **Small-county effect:** 5% of a small county is few customers, so small counties cross the threshold more easily (the positive rate is ~18% for mid-sized counties and falls to ~5% for the largest).
- **Zone-coded storms are blurry:** a zone report is assigned to every county the zone touches, so their outage rate (9.9%) is half that of county-coded storms (17.9%).
- **Denominator estimates** for North Carolina and 30 other counties rely on a national customers-per-person ratio.
- **Overlapping storms make rows dependent** (one storm system, many counties). Evaluate with resampling by `episode_id`.
- 2025 (test) has almost no tropical activity in the features (9 rows), so tropical performance can't be checked on test.

8. Feature dictionary

Statistics are from the training rows. Shares are fractions (0.05 = 5%). `retro_*` columns are **leakage**: they describe the outcome and must never be model inputs.

Column	Group	Unit	Description	Missing (train)	Mean	Median	Max
`fips`	id	text	5-digit county FIPS code (Census 2023)	0			

Column	Group	Unit	Description	Missing (train)	Mean	Median	Max
episode_id	id	integer	NOAA Storm Events EPISODE_ID (one storm system)	0			
state_abbr	id	text	State postal code (lower 48 + DC)	0			
fema_region	id	1-10	FEMA region of the county	0			
year	id	year	UTC year of t0; sets the split (train 2017-2023, validation 2024, test 2025)	0			
t0	id	UTC time	Begin time of the episode's first report in this county	0			
n_events	storm	count	Storm reports in this county during the episode (drought/heat excluded)	0	1.88	1.00	83.00
n_tstm_wind	storm	count	Reports of type Thunderstorm Wind	0	0.56	0.00	60.00
n_hail	storm	count	Reports of type Hail	0	0.26	0.00	57.00
n_tornado	storm	count	Reports of type Tornado or Funnel Cloud	0	0.0538	0.00	35.00
n_high_wind	storm	count	Reports of type High Wind or Strong Wind	0	0.20	0.00	83.00
n_flood	storm	count	Flash Flood, Flood, Heavy Rain, Coastal/Lakeshore Flood, Storm Surge/Tide, Debris Flow	0	0.27	0.00	68.00
n_winter	storm	count	Winter Weather, Winter Storm, Heavy Snow, Blizzard, Lake-Effect Snow, Sleet	0	0.38	0.00	35.00
n_ice_storm	storm	count	Reports of type Ice Storm	0	0.0084	0.00	12.00
n_tropical	storm	count	Tropical Storm, Hurricane (Typhoon), Tropical Depression	0	0.0123	0.00	10.00
n_cold	storm	count	Extreme Cold/Wind Chill, Cold/Wind Chill, Frost/Freeze	0	0.0626	0.00	12.00
n_lightning	storm	count	Reports of type Lightning	0	0.0097	0.00	8.00
n_wildfire	storm	count	Reports of type Wildfire	0	0.0196	0.00	7.00
n_other	storm	count	Any other event type (e.g. Dense Fog, Rip Current, Dust Storm)	0	0.0409	0.00	11.00
n_county_coded	storm	count	Reports NOAA located by county (CZ_TYPE = C)	0	1.14	1.00	68.00
n_zone_coded	storm	count	Reports issued for an NWS forecast zone, mapped to counties with the dated crosswalk	0	0.74	0.00	83.00
share_zone_coded	storm	0-1	n_zone_coded / n_events	0	0.49	0.00	1.00
any_partial_county	storm	boolean	At least one zone report whose zone covers only part of the county	0	0.22	0.00	1.00
max_wind_kts	storm	knots	Highest wind magnitude among the reports (measured or estimated); empty if none reported	59.0%	54.03	52.00	173
max_hail_in	storm	inches	Largest hail size among Hail reports; empty if no hail	87.2%	1.29	1.00	6.42
max_hurricane_cat	storm	0-5	Highest Saffir-Simpson category reported; almost always empty	99.9%	1.76	1.00	5.00

Column	Group	Unit	Description	Missing (train)	Mean	Median	Max
`county_episode_hours`	storm	hours	First report begin to last report end in this county	0	19.08	4.00	744
`episode_counties`	storm	count	Counties touched by the whole episode (storm-system size)	0	9.70	7.00	96.00
`log_episode_counties`	storm	log10	log10(episode_counties)	0	0.81	0.85	1.98
`episode_states`	storm	count	States touched by the whole episode	0	1.00	1.00	3.00
`episode_events`	storm	count	Distinct reports in the whole episode, all counties	0	15.46	10.00	251
`month`	time	1-12	Month of t0 (UTC)	0	5.71	5.00	12.00
`local_hour`	time	0-23	Hour of t0 in local standard time (state's main time zone)	0	12.30	14.00	23.00
`log_customers`	county	log10	log10 of customers served: EAGLE-I MCC, or Census population x 0.508 where MCC is broken (all NC + 30 counties)	0	4.33	4.29	6.58
`log_pop`	county	log10	log10 of Census 2023 county population; empty for Connecticut's 8 old counties	0.4%	4.63	4.58	6.99
`dqi`	county	0-100	EAGLE-I data quality index of the FEMA region (2018-2022 values; 2017 uses 2018, 2023+ use 2022)	0	78.19	79.90	94.04
`pct_customers_tracked`	county	0-1	Share of the state's customers EAGLE-I tracks (same year mapping as dqi)	0	0.87	0.90	1.00
`pre_peak_24h`	pre-storm	0-1	Worst share of customers out in the 24 h before the start hour; empty if the day before wasn't observed	3.3%	0.0085	0.0003	1.00
`pct_out_at_start`	pre-storm	0-1	Share of customers out in the hour before the start hour; empty if the day before wasn't observed	3.3%	0.0032	0.00	1.00
`hist_episodes`	history	count	County-episodes for this county in the training years (training rows: other training years only)	0	86.63	68.00	463
`hist_outage_rate`	history	0-1	County's past outage rate, shrunk toward the national training rate: (pos + 20 x p0) / (n + 20)	0	0.13	0.12	0.58
`hist_mean_peak`	history	0-1	County's past mean peak share out, shrunk the same way	0	0.0315	0.0248	0.27
`target`	target	0/1	1 if peak share out in hours 0-23 from the start hour is >= 5% of customers	0	0.14	0.00	1.00
`retro_peak_24h`	retro (leakage)	0-1	Peak share of customers out in the 24 h window (defines the target)	0	0.0335	0.0019	1.00
`retro_peak_customers_out`	retro (leakage)	customers	Peak customers out in the 24 h window	0	1,237	56.00	630,764
`retro_hour_of_peak`	retro (leakage)	0-23	Hour of the peak after the start hour (earliest if tied)	22.0%	7.52	5.00	23.00

9. Responsible use and maintenance

- **No personal data:** all records are county-level aggregates or public storm reports.
- **Leakage guard:** never use `retro_*` columns as inputs; never fit anything on `holdout/`. County history must be rebuilt from training years only if the split changes.
- **Rebuilding:** run notebooks 01 → 02 → 04 (~25 minutes); 06 regenerates every statistic in this card.
- **Updates:** a new year needs the next EAGLE-I and NOAA releases, a crosswalk version check, and a re-run of notebooks 02 and 04. Then bump the version above.

10. Citation

- Oak Ridge National Laboratory. *EAGLE-I power outage data, 2014–2025*. figshare, dataset 24237376. https://figshare.com/articles/dataset/24237376
- NOAA National Centers for Environmental Information. *Storm Events Database*. https://www.ncdc.noaa.gov/stormevents/